Advanced Network Security and Architectures

METI

Report Module 2 [EN]

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Group 3

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3.1 Introduction

This section describes the configuration and analysis of IPSec VPN using IKEv1 with digital certificates (PKI-based authentication using SCEP). The objective is to replace pre-shared keys with RSA signatures issued by a Certification Authority (CA).

3.2 Topology Overview

RA operates as the Certification Authority (CA) and NTP server. R1 and R2 establish an IPSec tunnel authenticated using RSA signatures and certificates obtained through SCEP enrollment.

3.3 PKI Installation Verification

The Certification Authority (CA) and router certificates were verified using IOS show commands.

3.3.1 CA Configuration and Certificate (RA)

The CA is configured on router RA and operates in automatic enrollment mode.

```
RA#show crypto pki server
Certificate Server myCA:
    Status: enabled
    State: enabled
    Server's configuration is locked  (enter "shut" to unlock it)
    Issuer name: cn=saarCA
    CA cert fingerprint: 46CB9C10 CC370311 8CFCB545 E4DD2F58
    Granting mode is: auto
    Last certificate issued serial number (hex): 1
    CA certificate expiration timer: 16:17:23 UTC May 14 2029
    CRL NextUpdate timer: 22:17:23 UTC May 15 2026
    Current primary storage dir: nvram:
    Database Level: Minimum - no cert data written to storage
```

The CA certificate installed on RA confirms that RA is acting as a trusted root authority:

```
RA#show crypto ca certificate
CA Certificate
  Status: Available
  Certificate Serial Number (hex): 01
  Certificate Usage: Signature
  Issuer:
    cn=saarCA
  Subject:
```

```
    cn=saarCA
  Validity Date:
    start date: 16:17:23 UTC May 15 2026
    end   date: 16:17:23 UTC May 14 2029
  Associated Trustpoints: myCA
  Storage: nvram:saarCA#1CA.cer
```

The detailed certificate view confirms RSA key usage and X.509 extensions:

```
RA#show crypto ca certificate verbose
CA Certificate
  Status: Available
  Version: 3
  Certificate Serial Number (hex): 01
  Certificate Usage: Signature
  Issuer:
    cn=saarCA
  Subject:
    cn=saarCA
  Validity Date:
    start date: 16:17:23 UTC May 15 2026
    end   date: 16:17:23 UTC May 14 2029
  Subject Key Info:
    Public Key Algorithm: rsaEncryption
    RSA Public Key: (1024 bit)
  Signature Algorithm: MD5 with RSA Encryption
  Fingerprint MD5: 46CB9C10 CC370311 8CFCB545 E4DD2F58
  Fingerprint SHA1: 0B6BF47E F0932FC1 54A33FA8 4C1F48D1 80083958
  X509v3 extensions:
    X509v3 Key Usage: 86000000
      Digital Signature
      Key Cert Sign
      CRL Signature
    X509v3 Subject Key ID: 42BD5391 0D07D957 6CC26630 AABA09E4 A93162E2
    X509v3 Basic Constraints:
        CA: TRUE
    X509v3 Authority Key ID: 42BD5391 0D07D957 6CC26630 AABA09E4 A93162E2
    Authority Info Access:
  Associated Trustpoints: myCA
  Storage: nvram:saarCA#1CA.cer
```

3.3.2 Router Certificates (R1)

R1 successfully enrolled with the CA and received both the CA certificate and its own identity certificate:

```
R1#show crypto pki certificates
```

```
Certificate
  Status: Available
  Certificate Serial Number (hex): 03
  Certificate Usage: General Purpose
  Issuer:
    cn=saarCA
  Subject:
    Name: R1
    Serial Number: 9SO8DYF8WAKSOLHEG3ZVL
    hostname=R1+serialNumber=9SO8DYF8WAKSOLHEG3ZVL
  Validity Date:
    start date: 16:33:16 UTC May 15 2026
    end   date: 16:33:16 UTC May 15 2027
  Associated Trustpoints: myTrustpoint
```

The CA certificate is also stored locally to validate the trust chain:

```
CA Certificate
  Status: Available
  Certificate Serial Number (hex): 01
  Certificate Usage: Signature
  Issuer:
    cn=saarCA
  Subject:
    cn=saarCA
  Validity Date:
    start date: 16:17:23 UTC May 15 2026
    end   date: 16:17:23 UTC May 14 2029
  Associated Trustpoints: myTrustpoint
```

3.3.3 RSA Public Keys (R1 and R2)

The RSA public keys generated on R1 and R2 are extracted using the following IOS command:

```
show crypto key mypub rsa
```

R2 RSA Public Key:

```
R2#show crypto key mypub rsa
% Key pair was generated at: 16:32:05 UTC May 15 2026
Key name: R2
Key type: RSA KEYS
 Storage Device: private-config
 Usage: General Purpose Key
 Key is not exportable.
 Key Data:
```

```
  30819F30 0D06092A 864886F7 0D010101 05000381 8D003081 89028181 00C337FE
  B0DAEE4D 444B0AF7 A19A73B3 2D2FCD5E 27E09095 3017E393 2F27F7AA 581894E6
  5E7F50B3 0454ACF2 FD1E8E24 F91125DD 6D611D93 611D70AD 36C59F6F EAEE003C
  7704B6DB 41D9A165 27252864 17170DA2 F38A68D4 46C91133 36DBCBC5 B09AD90E
  0F05531B F7ABB4A5 0B954C0F 389C2195 3DE377FF 7FE63FB3 B02DF189 35020301
  0001
% Key pair was generated at: 16:57:46 UTC May 16 2026
Key name: R2.server
Key type: RSA KEYS
Temporary key
 Usage: Encryption Key
 Key is not exportable.
 Key Data:
  307C300D 06092A86 4886F70D 01010105 00036B00 30680261 00A38B5F B33F29E4
  E5E30AFB 512084C9 0A89E015 BCA1569A C10C686B B79BD8EC C3C25CAD 6EE74C8F
  C891E5FF 1B042503 710EC940 65712523 377F2776 77CDC4C0 D7D61754 57C88063
  19BF191B F48130CC 327EF733 1F877D95 5CB74450 AEEF7E99 45020301 0001
```

R1 RSA Public Key:

```
R1#show crypto key mypub rsa
% Key pair was generated at: 16:31:43 UTC May 15 2026
Key name: R1
Key type: RSA KEYS
 Storage Device: private-config
 Usage: General Purpose Key
 Key is not exportable.
 Key Data:
  30819F30 0D06092A 864886F7 0D010101 05000381 8D003081 89028181 00DF87BB
  11BD33EE C4E9DF4A C2FF2CCB 2ECCEFF9 713AAA45 633ADE4A C188109E 88680D67
  FB0A2F20 67257007 79DB3616 75D5459A 4898DE69 E74E80BA 77C993E8 A4B70D41
  00D1D2FB E37BF0E3 89C8BEEE 9830712E 8A4C13DC E30C5C72 D72FEDF2 A221F6DB
  9A1A1B8F 9A69BC61 3700CA1B B66DB044 72F24609 E0C1FE8D 24D1B3DA DD020301
  0001
% Key pair was generated at: 16:57:42 UTC May 16 2026
Key name: R1.server
Key type: RSA KEYS
Temporary key
 Usage: Encryption Key
 Key is not exportable.
 Key Data:
  307C300D 06092A86 4886F70D 01010105 00036B00 30680261 0095CC14 73EE9BA8
  1B0A40A6 E3EE90C1 BC2FAADF 2D532270 160D9039 C19C40CA 6A4E54B5 BAB8892C
  D266AE9A 53210F58 876755F6 381C44B6 5D71F57F 35705BC5 3E8D9746 61833A6A
  DB5E740B 8A22A486 83EE7AFA ACAB3798 27D03FC2 7C6EA875 FD020301 0001
```

Both routers generate 1024-bit RSA key pairs used for digital signature authentication during IKEv1 Phase 1.

3.3.4 Summary

The outputs confirm that:

- RA is correctly configured as a trusted CA (myCA).
- R1 has successfully obtained a signed X.509 certificate from the CA.
- The certificate chain is self-signed at the CA level.
- RSA key pairs are generated locally on each router and are bound to X.509 certificates issued by the CA for IKEv1 authentication using digital signatures.
- Both R1 and R2 possess locally generated RSA key pairs used for authentication and certificate binding in IKEv1.

3.4 SCEP Enrollment Analysis

SCEP (Simple Certificate Enrollment Protocol) is used to automate certificate distribution between routers and the CA using HTTP.

The enrollment process consists of:

- `CA certificate request`
- `CA certificate response`
- `Router certificate request`
- `Router certificate issuance`

3.4.1 CA Certificate Retrieval

Initially, R1 requests the CA certificate using the SCEP `GetCACert` operation over HTTP.

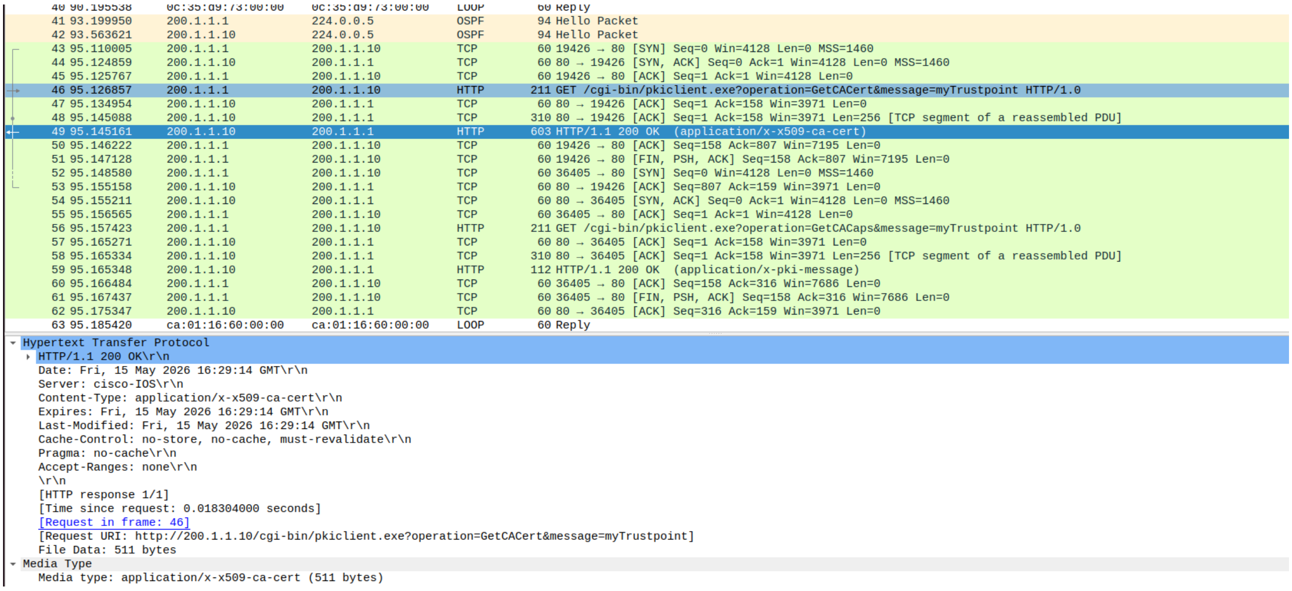

Figure 1: SCEP GetCACert request and CA certificate response

The CA replies with an HTTP `200 OK` message containing an X.509 CA certificate with media type `application/x-x509-ca-cert`. This certificate allows R1 to establish trust with the Certification Authority before certificate enrollment.

3.4.2 Certificate Enrollment Request

After establishing trust with the CA, R1 performs certificate enrollment using the SCEP `PKIOperation` request.

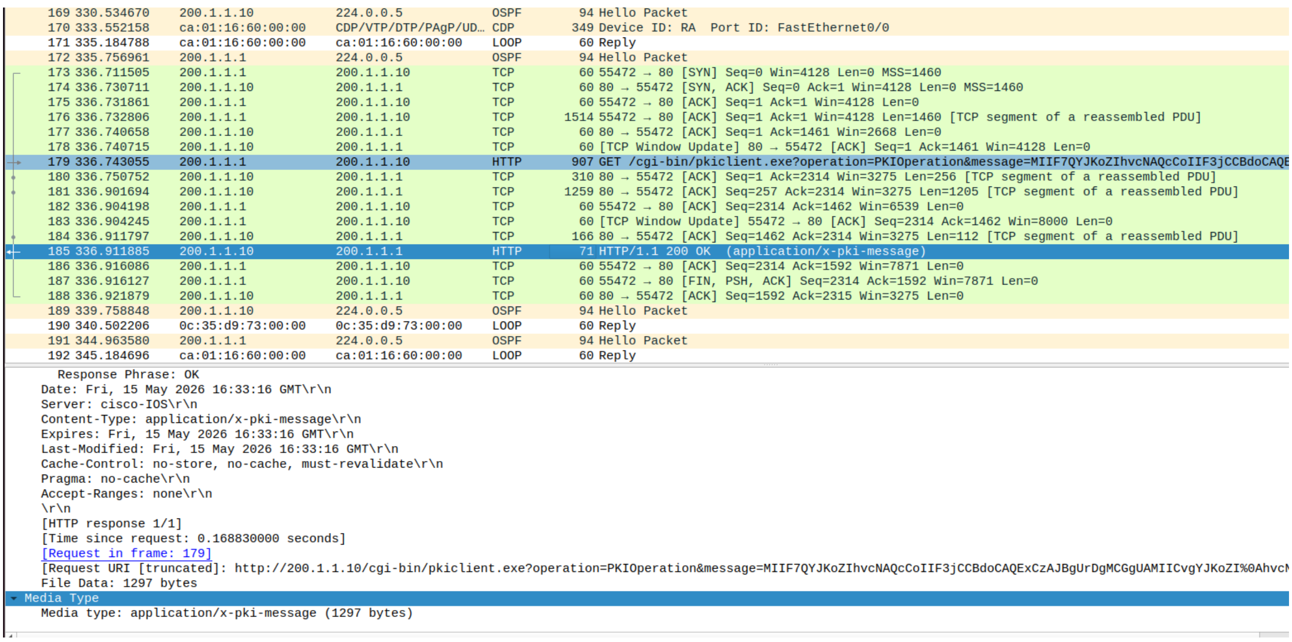

Figure 2: SCEP PKIOperation certificate enrollment request

The HTTP request contains a PKCS#10 certificate signing request (CSR) encoded in the message field. The CSR includes the router identity and its RSA public key, allowing the CA to generate a signed X.509 certificate for R1.

The captured HTTP messages show certificate exchange between R1 and the CA. The content-type field indicates certificate payload transfer (.crt).